SOFTWARE TEST PLAN

ONLINE QUIZ SYSTEM

PHASE 1 – VALIDATION SPECIFICATION | TEAM 13

Document Information

Item	Details
System	Online Quiz System
Department	CSE
Subject	Software Engineering
Semester / Section	5TH SEM, H
Team	13
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1. Introduction

The validation plan defines the testing activities that will be carried out to ensure that the Online Quiz System satisfies its specified requirements and behaves correctly under valid and invalid conditions. Since Phase 1 focuses on planning, the testing activities described below are planned for subsequent phases.

1.1 System Overview

The Online Quiz System allows a participant to start a quiz, view multiple-choice questions, select answers, submit the quiz and view the result. An administrator can log in and add, edit, delete and view questions.

2. Test Objectives

- Verify that the major functions of the Online Quiz System operate according to the specified requirements.
- Evaluate performance, usability, reliability and compatibility.
- Verify that restricted functionality is protected from unauthorized access.
- Verify that invalid, empty or incomplete inputs are handled correctly.

3. Test Scope and Test Approach

3.1 Functional Testing

Functional testing will verify that all major functions of the Online Quiz System operate according to the specified requirements.

- Starting a quiz
- Displaying multiple-choice questions
- Selecting answers
- Submitting a quiz
- Calculating the score

- Displaying quiz results
- Administrator login
- Adding questions
- Editing questions
- Deleting questions

3.2 Non-Functional Testing

Non-functional testing will evaluate the quality attributes of the system.

- Performance: Verify that quiz pages and questions load within an acceptable time.
- Usability: Verify that users can easily navigate through the quiz and understand the interface.
- Reliability: Verify that the system performs consistently without unexpected failures.
- Compatibility: Verify that the system functions correctly on supported browsers and devices.

3.3 Security Testing

Security testing will verify that the system protects restricted functionality and prevents unauthorized access.

- Administrator authentication
- Unauthorized access to administrator functions
- Invalid administrator login attempts
- User and administrator access control
- Session management
- Protection of quiz and administrator data

For example, if a normal quiz participant attempts to access an administrator page without authorization, the system should deny access.

3.4 Input Validation Testing

Input validation testing will verify that the system correctly handles invalid, empty, or incomplete inputs.

- Empty answer submission
- Empty question fields
- Missing answer options
- Invalid login credentials
- Invalid or unexpected input values
- Incomplete form submissions

The system should reject invalid inputs and display appropriate validation messages.

3.5 Usability Testing

Usability testing will verify that the Online Quiz System is simple and easy to use.

- Clear navigation
- Clearly labelled buttons
- Readable questions and answer options
- Simple quiz submission process
- Easy access to quiz results
- Clear and understandable error messages

A first-time user should be able to start, complete, and submit a quiz without requiring technical assistance.

4. Test Environment and Test Data

4.1 Hardware Requirements

Component	Requirement
Processor	Dual-core processor or above
RAM	Minimum 4 GB
Storage	At least 10 GB of free space
Network	Internet or local network connection
Device	Laptop, desktop, tablet or smartphone

4.2 Software Requirements

Component	Requirement
Operating System	Windows / Linux / macOS
Frontend	HTML, CSS and JavaScript
Backend	Node.js or suitable server-side technology
Database	MySQL / MongoDB
Browser	Chrome / Firefox / Edge / Safari
IDE	Visual Studio Code or any similar IDE

4.3 Test Data / Conditions

- Available quiz and multiple-choice questions.
- Submitted answers and correct answers stored in the system.
- Valid and incorrect administrator credentials.
- Valid, invalid, empty and incomplete question information.
- Authorized and unauthorized access attempts.

5. Test Cases and Validation Plan

The following test cases are taken from the test case summary in the source document. The cases cover participant functions, administrator functions, input validation and security.

Test Case ID	Test Case / Action	Expected Result
TC-01	Click on “Start Quiz”.	The quiz should start successfully.
TC-02	Start a quiz and view the questions.	Multiple-choice questions should be displayed.
TC-03	Select an answer for a question.	The selected answer should be recorded.
TC-04	Click on “Submit Quiz”.	The quiz should be submitted successfully.
TC-05	Submit a completed quiz.	The system should calculate the correct score.
TC-06	Complete the quiz.	The score/result should be displayed.
TC-07	Enter valid administrator credentials.	The administrator should be logged in.
TC-08	Add a new question.	The question should be added successfully.
TC-09	Edit an existing question.	The question should be updated successfully.
TC-10	Delete an existing question.	The question should be deleted successfully.
TC-11	Submit without providing a required answer.	The system should reject the input and display a validation message.
TC-12	Attempt to access administrator functions without authorization.	Access should be denied.
TC-13	Submit incomplete required information.	The system should reject the submission.
TC-14	Enter incorrect administrator login credentials.	Login should fail and an appropriate message should be displayed.

TC-15	Enter invalid input in a form field.	The system should reject or appropriately handle the input.

5.1 Security Validation

Security testing will verify that the system protects restricted functionality and prevents unauthorized access.

- Administrator authentication.
- Unauthorized access to administrator functions.
- Invalid administrator login attempts.
- User and administrator access control.
- Session management.
- Protection of quiz and administrator data.

5.2 Basic Penetration-Testing Plan

Basic penetration testing will be planned to identify common security vulnerabilities in the Online Quiz System. The testing will focus on authentication, authorization, input handling, and session management.

1. Unauthorized Access: Attempt to access administrator pages without logging in. Access should be denied.
2. Invalid Login Attempts: Attempt to log in using incorrect credentials. The system should reject the login attempt.
3. Input Manipulation: Enter unexpected or malicious input into form fields. The system should safely handle or reject such input.
4. Session Security: Attempt to access protected pages after logging out. Access should be denied.
5. Access Control: Verify that normal users cannot perform administrator operations.

5.3 Input Validation Testing

- Empty answer submission
- Empty question fields
- Missing answer options
- Invalid login credentials
- Invalid or unexpected input values
- Incomplete form submissions

5.4 Usability Testing

- Clear navigation
- Clearly labelled buttons
- Readable questions and answer options
- Simple quiz submission process
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5.5 Non-Functional Validation

- Performance: Verify that quiz pages and questions load within an acceptable time.
- Usability: Verify that users can easily navigate through the quiz and understand the interface.
- Reliability: Verify that the system performs consistently without unexpected failures.
- Compatibility: Verify that the system functions correctly on supported browsers and devices.

6. Requirements Traceability

The Requirements Traceability Matrix (RTM) establishes a relationship between system requirements, test cases, and validation methods. It ensures that every requirement can be verified and that no requirement is missed during testing.

Requirement ID	Requirement	Type	Test Case	Validation Method
FR-01	Start a quiz	FR	TC-01	Functional Testing
FR-02	Display multiple-choice questions	FR	TC-02	Functional Testing
FR-03	Select an answer	FR	TC-03	Functional Testing
FR-04	Navigate between questions	FR	TC-04	Functional Testing
FR-05	Submit the quiz	FR	TC-05	Functional Testing
FR-06	Calculate the user's score	FR	TC-06	Functional Testing
FR-07	Display the quiz result	FR	TC-06	Functional Testing
FR-09	Administrator login	FR	TC-07	Functional Testing
FR-10	Add questions	FR	TC-08	Functional Testing
FR-11	Edit questions	FR	TC-09	Functional Testing
FR-12	Delete questions	FR	TC-10	Functional Testing
VFR-01	Reject invalid or empty answer inputs	VFR	TC-11	Input Validation / Boundary Testing
VFR-03	Prevent unauthorized access to administrator functions	VFR	TC-12	Security Testing
VFR-05	Prevent submission of incomplete required information	VFR	TC-13	Input Validation
VFR-04	Reject invalid administrator login credentials	VFR	TC-14	Security Testing
VFR-10	Display appropriate error messages for invalid inputs	VFR	TC-15	Usability / Input Validation

7. Test Case Summary

Test Case ID	Test Case Description	Expected Result
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8. Testing Limitations Stated in Phase 1

Phase 1 is mainly concerned with planning and validation rather than carrying out actual penetration testing. The testing activities described in this document are planned for subsequent phases.